Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

APPLIED BUSINESS

Practical 12

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Practical 12: Analysis of Variance and Experimental Design

Part A: Analysis of Variance (ANOVA)

1. Load the Retail and werehouse Sale.csv dataset using Pandas.

Code:-

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as stats

# 1. Load the dataset using Pandas
file_path = '/content/retail_sales_dataset.csv'
df = pd.read_csv(file_path)
```

2. Display the first five records and identify the number of rows and columns.

Code:-

```
print("First 5 records:")
display(df.head())
print(f"\nDataset dimensions: {df.shape[0]} rows, {df.shape[1]
```

Output:-

First 5 records:

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
0	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
1	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
2	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
3	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
4	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100

Dataset dimensions: 1000 rows, 9 columns

3. Display the column names and data types.

Code:-

```
print("Columns and Data Types:")
display(df.dtypes)
```

Output:-

Columns and Data Types:

	0
Transaction ID	int64
Date	object
Customer ID	object
Gender	object
Age	int64
Product Category	object
Quantity	int64
Price per Unit	int64
Total Amount	int64

dtype: object

4. Check for missing values and duplicate records.

Code:-

```
missing_values = df.isnull().sum()
duplicates_count = df.duplicated().sum()

print("Missing values per column:")
print(missing_values)
print(f"\nNumber of duplicate records: {duplicates_count}")
```

```
item_type_col = 'Product Category'
retail_sales_col = 'Total Amount'

print(f"Mapped 'ITEM TYPE' to dataset column: '{item_type_col}'")
print(f"Mapped 'RETAIL SALES' to dataset column: '{retail_sales_col}'")
```

Output:-

```
Mapped 'ITEM TYPE' to dataset column: 'Product Category'
Mapped 'RETAIL SALES' to dataset column: 'Total Amount'
```

```
Missing values per column:
Transaction ID      0
Date                0
Customer ID         0
Gender              0
Age                 0
Product Category    0
Quantity            0
Price per Unit      0
Total Amount        0
dtype: int64

Number of duplicate records: 0
```

5. Display the different categories present in the ITEM TYPE column.

Code:-

```
unique_categories = df[item_type_col].unique()
print(f"Unique categories in '{item_type_col}':")
for cat in unique_categories:
    print(f" - {cat}")
```

Output:-

```
Unique categories in 'Product Category':
  - Beauty
  - Clothing
  - Electronics
```

6. Calculate the average RETAIL SALES for each ITEM TYPE.

Code:-

```
avg_sales = df.groupby(item_type_col)[retail_sales_col].mean().reset_index()
avg_sales.columns = [item_type_col, f'Average {retail_sales_col}']
print("Average Sales by Category:")
display(avg_sales)
```

Output:-

Average Sales by Category:

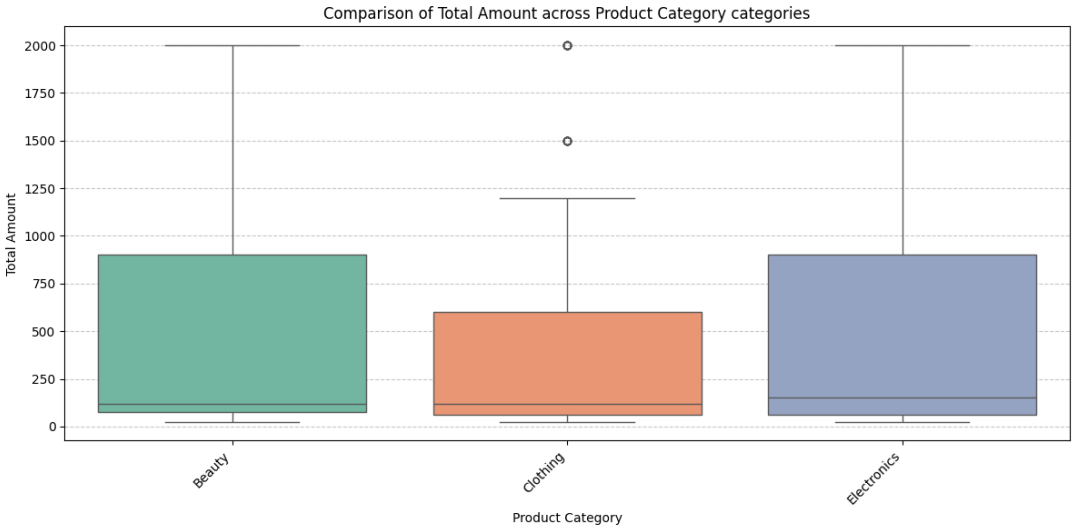
	Product Category	Average Total Amount
0	Beauty	467.475570
1	Clothing	443.247863
2	Electronics	458.786550

7. Create a box plot to compare RETAIL SALES across different ITEM TYPE categories.

Code:-

```
plt.figure(figsize=(12, 6))
sns.boxplot(data=df, x=item_type_col, y=retail_sales_col, palette='Set2')
plt.title(f'Comparison of {retail_sales_col} across {item_type_col} categories')
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

Output:-



8. Formulate the null and alternative hypotheses to determine whether the average retail sales differ among item types.

Output:-

- **Null Hypothesis (H_0):** The population mean of retail sales is the same across all item type categories. ($H_0 : \mu_1 = \mu_2 = \ldots = \mu_k$)
- **Alternative Hypothesis (H_1):** At least one item type has a different population mean of retail sales from the others.

9. Perform one-way ANOVA using ITEM TYPE as the factor and RETAIL SALES as the response variable.

Code:-

```
from scipy.stats import f_oneway

# Create groups based on Product Category (ITEM TYPE)
groups = [
    group["Total Amount"].dropna()
    for name, group in df.groupby("Product Category")
]

# Perform one-way ANOVA
f_statistic, p_val = f_oneway(*groups)

print("F-statistic:", f_statistic)
print("p-value:", p_val)
```

10. Calculate and display the F-statistic and p-value.

Output:-

```
F-statistic: 0.159430595139859
p-value: 0.8526508750301867
```

11. Compare the p-value with the significance level of 0.05.
12. State whether the null hypothesis is rejected or not rejected.
13. Write the conclusion about the difference in retail sales among different item types.

Code:-

```
# 11. Compare the p-value with the significance level of 0.05
# 12. State whether the null hypothesis is rejected or not rejected
```

```
# 13. Write the conclusion about the difference in retail sales among different item
types

significance_level = 0.05

print("--- Evaluation & Conclusion ---")
if p_value < significance_level:
    conclusion = (f"Since the p-value ({p_value:.4e}) is less than the significance
level of {significance_level}, "
                  f"we REJECT the null hypothesis. There is statistically significant
evidence to suggest that "
                  f"average retail sales differ among the different item types.")
else:
    conclusion = (f"Since the p-value ({p_value:.4e}) is greater than or equal to the
significance level of {significance_level}, "
                  f"we FAIL TO REJECT the null hypothesis. There is no statistically
significant evidence to suggest "
                  f"that average retail sales differ among the different item types.")
print(conclusion)
```

Output:-

```
--- Evaluation & Conclusion ---
Since the p-value (8.5265e-01) is greater than or equal to the significance level of 0.05, we FAIL TO REJECT t
```

Part B: Factorial Experiment

14. Select ITEM TYPE and YEAR as two factors and RETAIL SALES as the response variable.
15. Display the average retail sales for each combination of ITEM TYPE and YEAR.
16. Create a suitable visualization to compare retail sales for different item types across years.
17. Analyze the main effect of ITEM TYPE on retail sales.
18. Analyze the main effect of YEAR on retail sales.
19. Analyze the interaction effect between ITEM TYPE and YEAR.
20. Perform a suitable factorial ANOVA using the selected factors.
21. Display the statistical results and p-values for the main and interaction effects.
22. Identify whether ITEM TYPE, YEAR, or their interaction has a significant effect on RETAIL SALES.
23. Write the final business conclusion based on the ANOVA and factorial experiment Results.

Code:-

```
# ==========================================
# PART B: FACTORIAL EXPERIMENT
# ==========================================

# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from statsmodels.formula.api import ols
```

```
# ---------------------------------------------
# 14. Select ITEM TYPE and YEAR as factors
#     and RETAIL SALES as response variable
# ---------------------------------------------

# Load dataset
data = pd.read_csv("Retail and werehouse Sale.csv")

# Display selected columns
print("Selected Variables:")
print(data[["ITEM TYPE", "YEAR", "RETAIL SALES"]].head())


# ---------------------------------------------
# Data Cleaning
# ---------------------------------------------

# Remove rows with missing values in required columns
df = data[["ITEM TYPE", "YEAR", "RETAIL SALES"]].dropna()

# Convert ITEM TYPE and YEAR to categorical variables
df["ITEM TYPE"] = df["ITEM TYPE"].astype("category")
df["YEAR"] = df["YEAR"].astype("category")

print("\nCleaned Data:")
print(df.head())


# ---------------------------------------------
# 15. Average RETAIL SALES for each
#     combination of ITEM TYPE and YEAR
# ---------------------------------------------

average_sales = df.groupby(
    ["ITEM TYPE", "YEAR"],
    observed=True
)["RETAIL SALES"].mean().reset_index()

print("\nAverage Retail Sales for Each ITEM TYPE and YEAR:")
print(average_sales)


# ---------------------------------------------
# 16. Visualization
# ---------------------------------------------

plt.figure(figsize=(14, 7))

sns.lineplot(
    data=average_sales,
    x="YEAR",
    y="RETAIL SALES",
    hue="ITEM TYPE",
```

```
    marker="o"
)

plt.title("Average Retail Sales by Item Type Across Years")
plt.xlabel("Year")
plt.ylabel("Average Retail Sales")
plt.xticks(rotation=45)
plt.legend(title="Item Type", bbox_to_anchor=(1.05, 1), loc="upper left")
plt.tight_layout()
plt.show()


# ---------------------------------------------
# Another suitable visualization: Heatmap
# ---------------------------------------------

pivot_table = df.pivot_table(
    values="RETAIL SALES",
    index="ITEM TYPE",
    columns="YEAR",
    aggfunc="mean"
)

plt.figure(figsize=(14, 8))

sns.heatmap(
    pivot_table,
    annot=True,
    fmt=".1f"
)

plt.title("Average Retail Sales: Item Type vs Year")
plt.xlabel("Year")
plt.ylabel("Item Type")
plt.tight_layout()
plt.show()


# ---------------------------------------------
# 17, 18, 19 & 20
# Factorial ANOVA
# ---------------------------------------------

# Create the two-way factorial ANOVA model
model = ols(
    "Q('RETAIL SALES') ~ C(Q('ITEM TYPE')) * C(Q('YEAR'))",
    data=df
).fit()

# Perform ANOVA
anova_table = sm.stats.anova_lm(model, typ=2)

print("\n=============================================")
print("TWO-WAY FACTORIAL ANOVA RESULTS")
```

```
print("=============================================")

print(anova_table)


# ---------------------------------------------
# 21. Display statistical results and p-values
# ---------------------------------------------

print("\n=============================================")
print("F-STATISTICS AND P-VALUES")
print("=============================================")

print("ANOVA Results:")
print(anova_table[["F", "PR(>F)"]])


# ---------------------------------------------
# 17. Main Effect of ITEM TYPE
# ---------------------------------------------

item_type_p = anova_table.loc[
    "C(Q('ITEM TYPE'))",
    "PR(>F)"
]

item_type_f = anova_table.loc[
    "C(Q('ITEM TYPE'))",
    "F"
]

print("\n=============================================")
print("17. MAIN EFFECT OF ITEM TYPE")
print("=============================================")

print("F-statistic:", item_type_f)
print("p-value:", item_type_p)

if item_type_p < 0.05:
    print("ITEM TYPE has a significant effect on RETAIL SALES.")
else:
    print("ITEM TYPE does not have a significant effect on RETAIL SALES.")


# ---------------------------------------------
# 18. Main Effect of YEAR
# ---------------------------------------------

year_p = anova_table.loc[
    "C(Q('YEAR'))",
    "PR(>F)"
]

year_f = anova_table.loc[
```

```
    "C(Q('YEAR'))",
    "F"
]

print("\n=============================================")
print("18. MAIN EFFECT OF YEAR")
print("=============================================")

print("F-statistic:", year_f)
print("p-value:", year_p)

if year_p < 0.05:
    print("YEAR has a significant effect on RETAIL SALES.")
else:
    print("YEAR does not have a significant effect on RETAIL SALES.")


# ---------------------------------------------
# 19. Interaction Effect
# ---------------------------------------------

interaction_p = anova_table.loc[
    "C(Q('ITEM TYPE')):C(Q('YEAR'))",
    "PR(>F)"
]

interaction_f = anova_table.loc[
    "C(Q('ITEM TYPE')):C(Q('YEAR'))",
    "F"
]

print("\n=============================================")
print("19. INTERACTION EFFECT")
print("=============================================")

print("F-statistic:", interaction_f)
print("p-value:", interaction_p)

if interaction_p < 0.05:
    print("There is a significant interaction between ITEM TYPE and YEAR.")
else:
    print("There is no significant interaction between ITEM TYPE and YEAR.")


# ---------------------------------------------
# 22. Identify significant effects
# ---------------------------------------------

print("\n=============================================")
print("22. SIGNIFICANCE TEST")
print("=============================================")

alpha = 0.05
```

```
print("Significance Level =", alpha)

if item_type_p < alpha:
    print("ITEM TYPE: Significant")
else:
    print("ITEM TYPE: Not Significant")

if year_p < alpha:
    print("YEAR: Significant")
else:
    print("YEAR: Not Significant")

if interaction_p < alpha:
    print("ITEM TYPE × YEAR: Significant")
else:
    print("ITEM TYPE × YEAR: Not Significant")


# ---------------------------------------------
# 23. Final Business Conclusion
# ---------------------------------------------

print("\n=============================================")
print("23. FINAL BUSINESS CONCLUSION")
print("=============================================")

if item_type_p < alpha:
    item_result = "ITEM TYPE has a significant effect on retail sales."
else:
    item_result = "ITEM TYPE does not have a significant effect on retail sales."

if year_p < alpha:
    year_result = "YEAR has a significant effect on retail sales."
else:
    year_result = "YEAR does not have a significant effect on retail sales."

if interaction_p < alpha:
    interaction_result = (
        "There is a significant interaction between ITEM TYPE and YEAR, "
        "meaning the effect of item type on retail sales changes across years."
    )
else:
    interaction_result = (
        "There is no significant interaction between ITEM TYPE and YEAR, "
        "meaning there is not enough evidence that the effect of item type "
        "changes across years."
    )

print(item_result)
print(year_result)
print(interaction_result)

print("\nOverall, the factorial ANOVA helps identify whether differences "
      "in retail sales are associated with item type, year, and their "
```

"combined interaction.")

Output:-

```
Selected Variables (Product Category, Quarter, Total Amount) - Cleaned:
  Product Category Quarter  Total Amount
0           Beauty  2023Q4           150
1         Clothing  2023Q1          1000
2      Electronics  2023Q1            30
3         Clothing  2023Q2           500
4           Beauty  2023Q2           100

Average Retail Sales for each Product Category and Quarter:
   Product Category Quarter  Total Amount
0           Beauty  2023Q1    514.027778
1           Beauty  2023Q2    431.097561
2           Beauty  2023Q3    453.521127
3           Beauty  2023Q4    462.407407
4         Clothing  2023Q1    440.721649
5         Clothing  2023Q2    411.534653
6         Clothing  2023Q3    432.112676
7         Clothing  2023Q4    494.939024
8       Electronics  2023Q1    442.153846
9       Electronics  2023Q2    550.823529
```


```
Selected Variables (Product Category, Quarter, Total Amount) - Cleaned:
  Product Category Quarter  Total Amount
0           Beauty  2023Q4           150
1         Clothing  2023Q1          1000
2      Electronics  2023Q1            30
3         Clothing  2023Q2           500
4           Beauty  2023Q2           100

Average Retail Sales for each Product Category and Quarter:
   Product Category Quarter  Total Amount
0           Beauty  2023Q1    514.027778
1           Beauty  2023Q2    431.097561
2           Beauty  2023Q3    453.521127
3           Beauty  2023Q4    462.407407
4         Clothing  2023Q1    440.721649
5         Clothing  2023Q2    411.534653
6         Clothing  2023Q3    432.112676
7         Clothing  2023Q4    494.939024
8       Electronics  2023Q1    442.153846
9       Electronics  2023Q2    550.823529
```


```
pivot_table = df.pivot_table(
```



